

MATHEMATICS DAYS IN SOFIA 2026

Knowledge Before Action

Hello, my name is Aleks

Introduction

- Bachelors in
  Computer Science in the Technical University in Sofia
- Masters in
  Information Technology for Business Management
- PHD candidate in
  Bulgarian Academy of Sciences with a focus on AI systems
- Hobbies:
  - Latin dancing – Bachata, Salsa
  - Building AI Powered products and helping the world run best



What does it take for an AI agent to act reliably?

Central question · Four parts ahead

- **The Problem** — why information systems fragment and resist automation
- **Why Agents Fail** — architectural failures, not language-model failures
- **The Framework** — a multi-layer knowledge-before-action architecture
- **In Practice** — illustrative domain, ablation design, contributions



Part 1

The Problem

Modern Organisations: Three Structural Problems



- **Fragmented** — ERP, CRM, SaaS, portals in isolation
- **Heterogeneous** — different interfaces, schemas, protocols
- **Passive** — systems store knowledge but cannot reason over it
- **Extracting knowledge and acting on it requires humans to do all the translation.**

Fragmented business information systems — no shared schema, no unified access

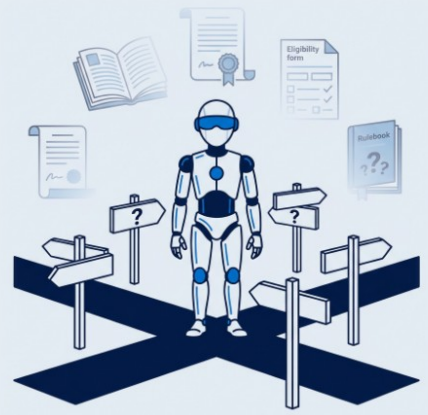
Part 2

Why Current AI Agents Fail

Two Root Causes — Both Architectural

Domain Ignorance

- Selects the *wrong procedure*
- Ignores eligibility rules
- Hallucinates regulatory codes
- No authoritative domain state to check against



Action Without Grounding

- Acts on *parametric* (trained) knowledge
- Not the current authoritative domain state
- These are **architectural** failures
- Not language-model quality failures



"The agent acts before it knows."

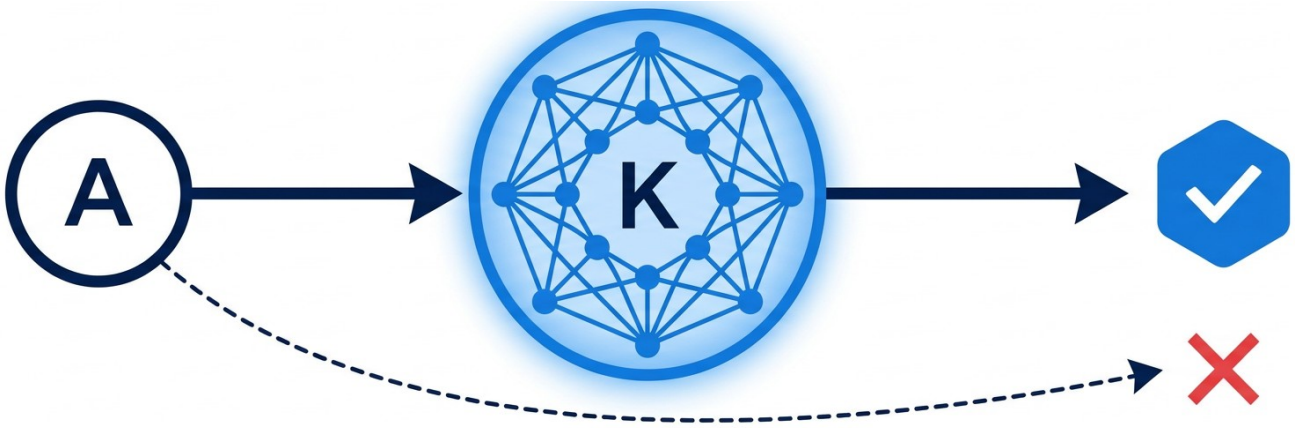
— The core diagnosis



The Testable Hypothesis

Mathematical Foundations of CS and AI · IMI-BAS 2026

- Let **A** be an autonomous agent operating over information system **I**
- Let **K** be a structured knowledge base encoding domain semantics of **I**
- **Claim:** grounding A's action selection in K
  - reduces task error compared to A operating without K
- *This is the testable hypothesis the framework is built to instantiate.*

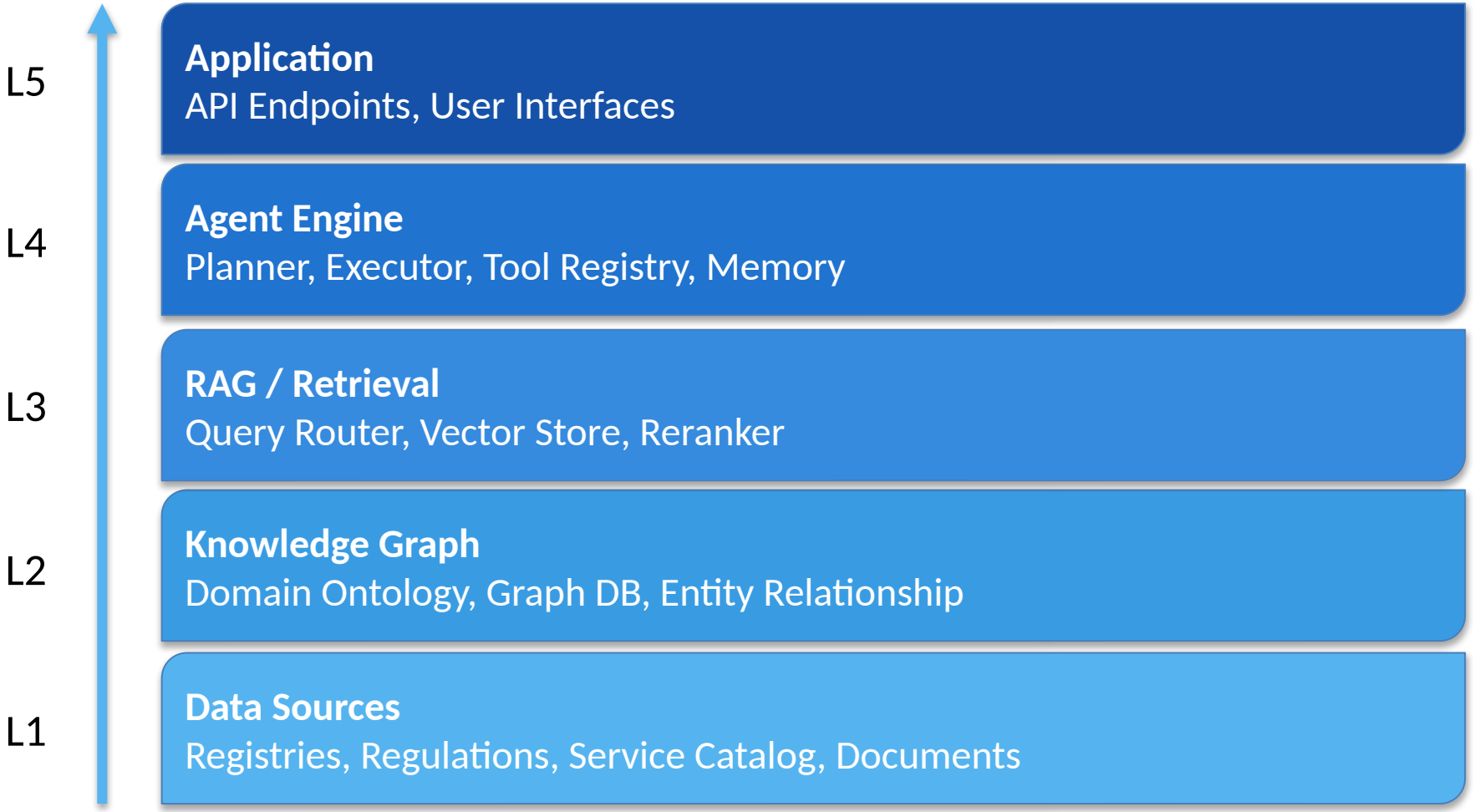


Part 3

The Proposed Framework

A 5-Layer Knowledge-Before-Action Architecture

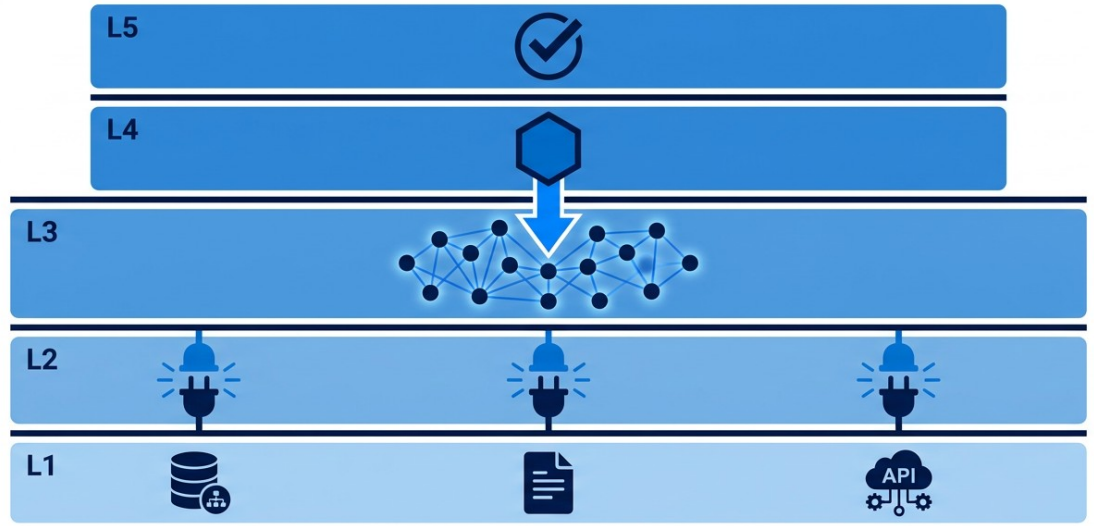
System-agnostic · Each layer independently replaceable · Contracts between layers



What Makes This Architecture Different

Four key properties revealed in sequence

- **One responsibility per layer** — independently replaceable
- **REST/gRPC contracts** between layers — no framework coupling
- **The agent (L4) cannot act** without first querying L2 and L3
  - This constraint is *structural*, not a guideline or convention
- **One adapter per source type** — never hardcoded integration
- **Governance by design** — audit, access control, explainability are structural



Five Core Design Principles

Each principle addresses a known failure mode in current AI deployments

Separation of Concerns

Each layer has one responsibility
Independently replaceable

Technology Agnosticism

Layers communicate via contracts
No framework coupling

Governance by Design

Audit, access control, explainability
are structural — not optional

Graceful Degradation

If KG unavailable, fall back
to retrieval gracefully

Progressive Autonomy

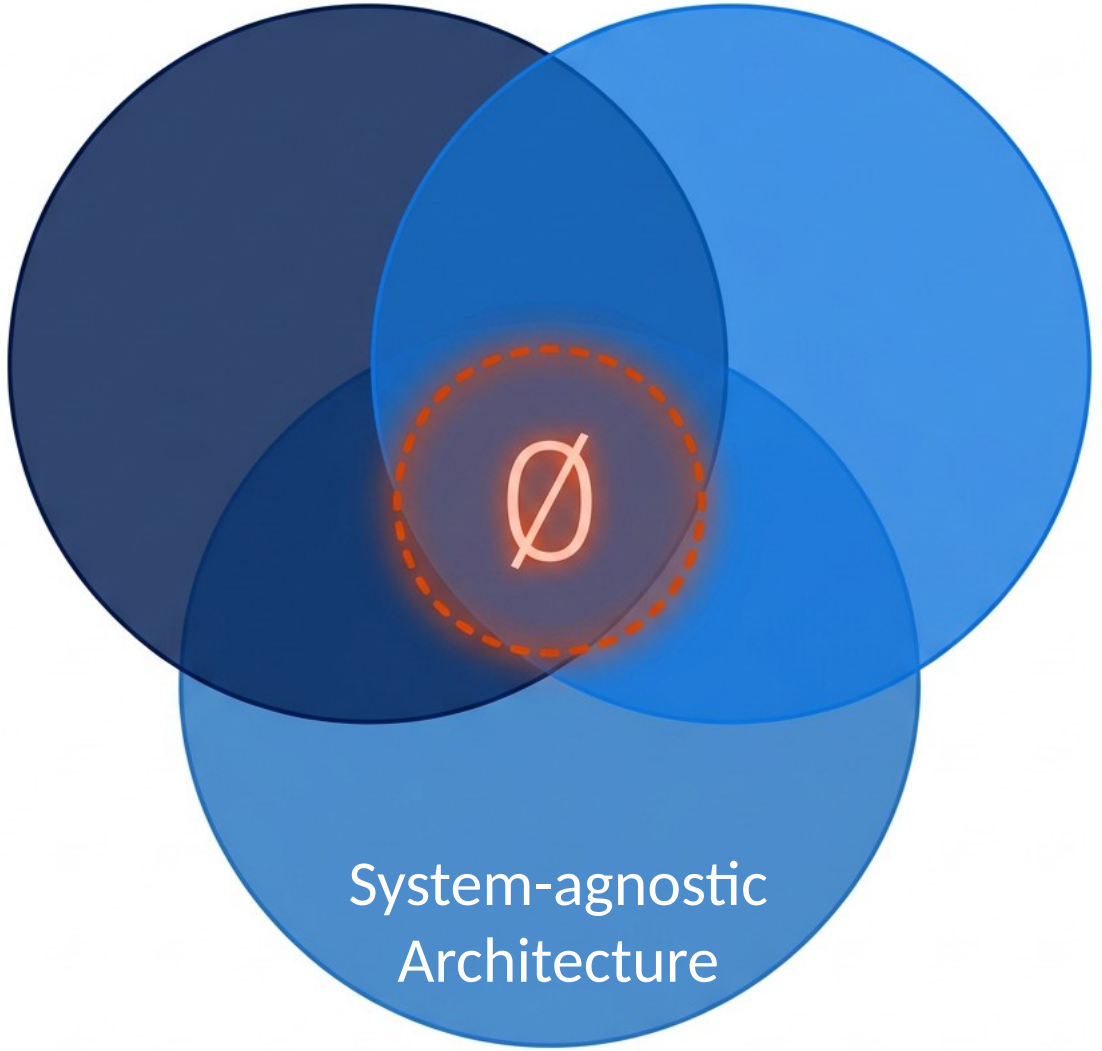
Routine tasks fully autonomous
High-stakes require human approval

Framework’s Place in Space

Combining 3 crucial elements

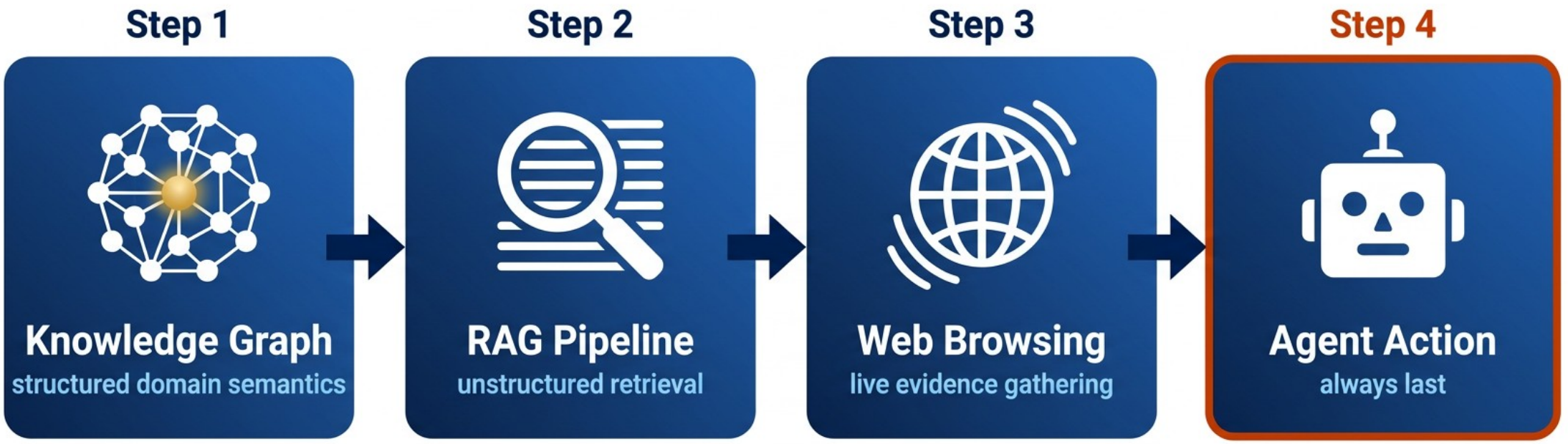
Action-capable Agents

Knowledge from Authoritative Sources



Knowledge Before Action — The Pipeline

The agent is always last · Every action has a knowledge provenance



Integrated Knowledge + Contextual Retrieval + Real-time Data = Intelligent Agent Action

The synergistic framework for advanced AI systems, ensuring robust and accurate decision-making.

Part 4

In Practice

Illustrative Domain: Bulgarian Citizen Service Delivery

Why This Domain

- Publicly available government data (egov.bg)
- Real regulatory complexity
  - Eligibility rules, required documents, fees
- Multilingual context (Bulgarian + EU regulations)
- Representative of the broader class
  - Fragmented, heterogeneous, passive systems

What the Agent Actually Does

- Citizen request arrives
- **KG lookup + RAG retrieval** — get verified values
- **Agent plans** the steps for this citizen's case
- **Navigates** real egov.bg web interfaces
- **Fills** real forms with KG-verified values

- *The agent acts. It does not just answer questions.*

Planned Evaluation: Factorial Ablation Design

Each condition removes one component — isolating its causal contribution

- **Full System** (KG + RAG + Agent) — the complete framework
- **Agent + RAG** (no KG) — isolates the KG contribution
- **Agent + KG** (no RAG) — isolates the RAG contribution
- **Baseline** (no KG, no RAG). — parametric knowledge only

- Metrics: *task completion rate · form-filling accuracy*
- Design question: how much of agent performance is attributable to structured knowledge?



Summary

What This Research Contributes

Three Core Contributions

Open question: Under what conditions does structured knowledge become a bottleneck — when it is stale, incomplete, or contradicted by retrieved documents?

1. **Architectural Framework** —
   System-agnostic multi-layer framework for integrating AI agents with business information systems
2. **Formal Design Principle** —
   Knowledge before action as a structural architectural invariant, not a guideline
3. **Evaluation Methodology** —
   Factorial ablation design attributing agent performance to specific knowledge components



Thank You

∀ action a : a is grounded in K · knowledge before action

Aleksandar Aytov · IMI-BAS · Mathematics Days in Sofia 2026